Object Oriented Programming

Assignment # 1 (CLO-2)

Complex Computing Problem (CCP)

OOP Assignment: Crisis-Based Online University System

Scenario

Due to ongoing global conflicts and a severe fuel crisis, the Government of Punjab has imposed a partial lockdown across the province. As a result, all universities are required to shift to online operations.

Namal University has transitioned to a fully online system to ensure continuity of academic and administrative activities.

To manage this sudden shift, the university plans to develop a **Crisis-Based Online University Management System** that digitally handles:

- Online classes and sessions
- Student participation
- Faculty management
- Resource allocation (recordings, attendance logs, etc.)
- Emergency adjustments (network issues, rescheduling, etc.)

However, due to urgency, the university has hired **you as a system designer**, and instead of a fully working system, they require a **conceptual OOP-based design** that demonstrates strong understanding of object-oriented principles.

Assignment Requirements

Part 1: Object Identification & Design

You are required to:

1. Identify **at least 4–5 major classes** from the scenario
2. For each class:
    - Define **data members**
    - Define **member functions**
    - Clearly justify your design choices
3. Identify relationships:
    - **Composition (must be included)**

Important:
Your design should not be obvious - justify **why** a class exists and **why** attributes belong to it.

Part 2: Class Diagram

- Draw a **detailed UML Class Diagram**
- Clearly show:
  - Classes
  - Attributes & functions
  - Relationships
  - **Composition (must be properly labeled)**

Part 3: Implementation (Conceptual C++ Design)

You are required to **write C++-like code** (compilation is NOT required).

Your design must include:

Required Concepts

Concept	Requirement
Classes & Objects	Use meaningful real-world entities
Constructors	Parameterized constructors
Destructors	Proper use (even if conceptual)
Constant Member Functions	At least one per class
Object Composition	Mandatory (e.g., system contains sessions, sessions contain participants, etc.)
Shallow vs Deep Copy	Demonstrate both and explain difference
Operator Overloading	Use creatively (see twist below)

Part 4: Critical Thinking & Conceptual Traps

This section is the **core differentiator**.

Operator Overloading

You must:

- Attempt to overload **at least one operator that CANNOT be overloaded in C++** (e.g., ., ::, ?:, sizeof)

Then:

- Explain **why it cannot be overloaded**
- Provide an **alternative design approach**

Deep Copy vs Shallow Copy

- Create a situation where:
  - Shallow copy causes a problem
  - Deep copy resolves it
- Clearly explain:
  - Memory behavior
  - Consequences in your system

Design Justification

For each major decision:

- Why composition instead of inheritance?
- Why dynamic memory (if used)?
- Why a function is marked const?

Part 5: Simulation / Working Explanation

Instead of running code, you must:

- Explain step-by-step:
  - How objects are created
  - How they interact
  - What happens during copying, deletion, etc.

Think of this as a **dry run of your system**

Important Instructions

- Code **does NOT need to compile**
- Focus is on:
  - Design clarity
  - Concept application
  - Justification
- Avoid generic answers - be specific to the scenario

- Clearly label all sections

Submission Guidelines

- The submission deadline for this assignment is **Friday, 10-04-2026 at 11:59 PM**.
- Each group must submit:
  - The **code files**, and
  - A **comprehensive and well-documented report** explaining the design, implementation, and working of the system.
- This assignment is to be completed in **groups of exactly two (2) students**.
  - Individual submissions are **not allowed**.
  - Submissions with fewer than two members will **not be accepted under any circumstances**.
  - Students are responsible for forming their groups independently.
- The submitted document must be **detailed, well-structured, and clearly written**, demonstrating thorough understanding and explanation of all required concepts.